



OpenCoven

GitHub App Setup Guide

Configure, transfer, and verify the Coven GitHub App

Operator guide for creating the app, installing it on target repositories, wiring webhook configuration, and validating event delivery without exposing secrets.

Source context: [docs/app-manifest.json](#), [docs/self-hosting.md](#), [README.md](#), and [HOSTED.md](#)

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Purpose and Audience

This guide documents the browser setup and operational handoff for the OpenCoven GitHub App. It is written for maintainers or deployment operators who can create or transfer a GitHub App, install it on repositories, and configure the webhook service that receives GitHub events.

Use the placeholders exactly as placeholders until values are placed in secret storage. Do not paste real private keys, webhook secrets, installation tokens, personal tokens, or private email addresses into this document.

Operating model

The app listens for GitHub issue, comment, review, check, and pull request review events. It uses GitHub App installation tokens rather than user credentials, then invokes the configured Coven runtime or webhook deployment for the requested repository workflow.

Prerequisites

Prerequisite	What to confirm
GitHub account or organization permissions	You can create GitHub Apps under the intended owner or a temporary owner and can transfer ownership to OpenCoven/Coven when needed.
Public HTTPS webhook endpoint	A reachable endpoint such as <code>https://<public-host>/webhook</code> . For local testing, use a secure tunnel such as ngrok.
Webhook deployment access	You can deploy, restart, and inspect logs for <code>coven-github</code> or <code>coven-github-webhook</code> .
Secret storage location	You have a secure place for App ID, webhook secret, private key, and installation ID. Examples include a deployment secret manager or local untracked config.
Target repository access	You can install the GitHub App on the target organization, repository set, or selected repositories.

Setup Sequence

Follow these steps in order. Use the manifest flow for the fastest match to the repository defaults, or manual setup if the GitHub UI or organization policy requires it.

#	Step	Operator action and validation
1	Open GitHub app setup	Go to GitHub Developer Settings for a personal owner or the organization Settings > Developer settings > GitHub Apps page for an org-owned app.
2	Choose setup path	Use the manifest flow from <code>docs/app-manifest.json</code> when possible. Otherwise select New GitHub App and enter the settings manually.
3	Set core fields	Use an app name appropriate for the owner, homepage URL <code>https://opencoven.ai</code> or the deployment homepage, webhook URL <code>https://<public-host>/webhook</code> , redirect URL <code>https://<public-host>/app/created</code> , and private visibility unless a public app is explicitly intended.
4	Configure permissions	Set Contents: Read and write, Issues: Read and write, Pull requests: Read and write, Checks: Read and write, and Metadata: Read-only.
5	Subscribe to events	Subscribe to issues, issue_comment, pull_request_review, pull_request_review_comment, check_suite, and check_run.
6	Create the app	Submit the manifest or manual form. Confirm GitHub displays the newly created app settings page.
7	Generate private key	Generate and download the PEM private key. Store it outside source control and restrict access.
8	Record app values	Record the App ID and webhook secret in the deployment secret store. Do not paste the real private key into documentation or tickets.
9	Install on targets	Install the app on the OpenCoven/Coven organization or selected repositories that should receive automation.

10	Capture installation ID	Open the installation URL or inspect deployment logs after a webhook event to capture <installation-id>. Store it with the app configuration.
11	Configure runtime or webhook	For coven-github, set github.app_id, github.private_key_path, github.webhook_secret, and any runtime paths in config/local.toml. For coven-github-webhook, set equivalent environment secrets in the deployment.
12	Verify GitHub delivery	In the app settings, open Advanced > Recent Deliveries and confirm the ping event returns a 2xx response.
13	Trigger test flow	Create or update a test issue, label, assignment, or comment expected to wake the familiar. Confirm logs, webhook receipt, and the expected repository response.
14	Transfer ownership if needed	If the app was created under a temporary owner, transfer the GitHub App to OpenCoven/Coven through GitHub App settings and complete organization acceptance.
15	Re-check after transfer	After transfer, confirm webhook URL, secret, permissions, subscribed events, installations, recent delivery status, and saved App ID or slug references.

Manifest Defaults

The local manifest defines the intended baseline. Replace host placeholders before creating the app.

Field	Value or baseline
Homepage URL	https://opencoven.ai
Webhook URL	https://<public-host>/webhook
Redirect URL	https://<public-host>/app/created
Visibility	Private by default
Permissions	contents write; issues write; pull_requests write; checks write; metadata read
Webhook events	issues; issue_comment; pull_request_review; pull_request_review_comment; check_suite; check_run

Values to Save

Value	Where it comes from	Placeholder example	Store in
GitHub App ID	GitHub App settings page	<github-app-id>	Deployment secret or config value
Webhook secret	Manifest result or manually generated secret	<webhook-secret>	Secret manager, never source control
Private key PEM	Generated from GitHub App settings	<app-name>.private-key.pem	Restricted file path or secret mount
Installation ID	App installation URL, API, or logs after install	<installation-id>	Deployment config or secret
Webhook URL	Public deployment or tunnel	https://<public-host>/webhook	GitHub App settings and deployment notes
Redirect URL	Creation callback endpoint	https://<public-host>/app/created	GitHub App settings

Configuration Targets

Target	Configuration notes
coven-github	Use config/local.toml or equivalent deployment config for github.app_id, github.private_key_path, github.webhook_secret, worker.coven_code_bin, familiar settings, and optional server.cave_base_url.
coven-github-webhook	Use deployment environment secrets for the same GitHub App values. Keep app owner, slug, and installed repositories aligned.

Security Notes

Rule	Reason
Never commit PEM/private keys	The private key can mint installation tokens for the app and must stay outside git history.
Keep webhook secret in secret storage	GitHub webhook signatures depend on this value. If exposed, rotate it immediately.
Use installation tokens, not user credentials	GitHub App installation tokens limit access to the installed owner and repositories.
Rotate exposed secrets	Rotate webhook secrets, private keys, and any affected deployment credentials after accidental disclosure.
Restrict repository installation scope	Install on selected repositories unless the operator intentionally needs org-wide access.

Troubleshooting

Symptom	Likely cause	Resolution
Webhook 401, 403, or signature failure	Webhook secret mismatch, missing signature header, wrong endpoint, or stale deployment secret.	Confirm the GitHub App webhook secret matches the deployment secret. Restart the service after changing secrets and replay the GitHub delivery.
Missing permission error	The app lacks one of contents, issues, pull_requests, checks, or metadata permissions.	Update app permissions, accept any installation permission update prompt, then retry the flow.
App installed on wrong repositories	Installation scope selected the wrong owner or repository set.	Open Install App settings and adjust repository access. Re-check installation ID if the installation changed.
No webhook deliveries	Webhook disabled, URL unreachable, app not installed, event not subscribed, or repository action did not match subscribed events.	Confirm the webhook is active, endpoint is public HTTPS, event subscriptions include the tested action, and Recent Deliveries shows a ping.
Transfer-related regression	Ownership transfer changed app slug, installation state, permission acceptance, or operator assumptions.	After transfer, re-check webhook URL, secret, subscribed events, permissions, installations, app ID, and deployment config references.

Final Validation Checklist

Check	Done when
App exists under intended owner	The OpenCoven/Coven organization owns the app or has accepted the transfer.
Settings match manifest	URLs, visibility, permissions, and events match the intended baseline.
Secrets are stored safely	No real secret appears in git, docs, tickets, screenshots, or chat logs.
App is installed on targets	Selected repositories or organization access is correct.
Ping succeeds	Advanced > Recent Deliveries shows a 2xx ping response.
Test repository flow succeeds	A test issue, label, assignment, or comment triggers expected Coven behavior.
Transfer re-check complete	Post-transfer owner, installation, webhook, secret, and permission checks are complete.